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SOLID TUMOURS IN CHILDREN

Neuroblastoma

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- m/c intra-abdominal tumor in children.
- m/c extra-cranial solid tumor in children.
- 2nd m/c solid tumors in children (m/c : Intracranial/brain tumors)
- m/c malignancy in infancy.

Site of involvement :

Cell of origin : **Neuroblasts** (derived from neural crest cells).

Organs Involved :

1. Adrenal gland (m/c) → AKA suprarenal tumor.
2. Paravertebral sympathetic chain/ganglia (2nd m/c).
3. Posterior mediastinum.
4. Pelvis.

Pathology :

Gross : Solid tumor

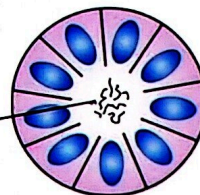
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microscopy : Small, round, uniform blue cells

(Scanty cytoplasm + hyperchromatic nuclei)

Forms **Homer-Wright rosettes** :

Rose petal/rosette appearance with central eosinophilic material called **neuropil** (cytoplasmic processes)



Note : D/d for small, round, uniform blue cells

- Non-Hodgkin's lymphoma.
- Rhabdomyosarcoma.
- Ewing's tumor.
- Neuroblastoma.

Clinical features :

Abdominal mass : Firm, fixed, irregular, midline .

Constitutional symptoms : Fever, Weight loss, Irritability.

Horner's syndrome : Anhidrosis, miosis & ptosis.

Heterochromia iridis : Variation in color of the two irises.

Features d/t metastasis :

- To **bone** (m/c site) → Bony pain.
- To **retro-orbital region** with infiltration → peri-orbital swelling and ecchymosis (**Raccoon eyes**)
- To skin → **Blueberry muffin lesions** (Purplish/bluish subcutaneous nodule)



Blueberry muffin lesions



Raccoon eyes

Paraneoplastic manifestations :

- Opsoclonus-myoclonus ataxia

Also dancing eye/dancing feet syndrome :

- Associated with anti-neuronal antibodies.

- VIP (Vasoactive intestinal polypeptide) hypersecretion → Intractable watery diarrhea, hypokalemia.

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Investigations :

Urine markers (Initial Investigation) :

- metabolites of catecholamines (Non specific)
 - VMA (vanillylmandelic acid).
 - HVA (homovanillic acid).
- NSE (Neuron specific enolase) : most Specific.

Imaging studies (Diagnostic investigations) :

- CT abdomen (m/c) :
 - Stippled calcifications.
 - Tumor mass displacing kidney inferolaterally (Not seen in Wilms's tumor)
- Radionuclide scan with ^{123}I MIBG (meta Iodo Benzyl guanidine) } To rule out metastases.
- PET-CT scan

Staging :

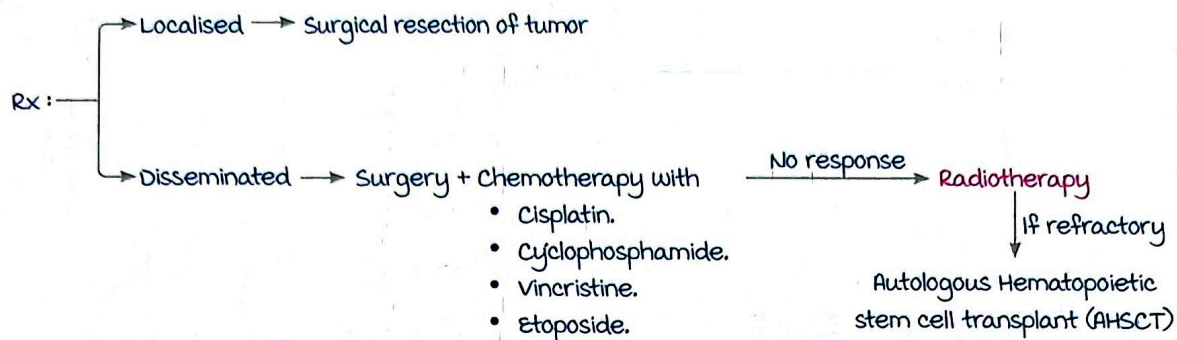
International neuroblastoma staging system (INSS) :

Stage I	Completely resectable tumors.
Stage 2	Incompletely resectable tumors. 2a : No lymph node involvement. 2b : With lymph node involvement (only ipsilateral).
Stage 3	• Crosses the midline (or) • Unilateral tumor + contralateral lymph node involvement.
Stage 4	• Disseminated tumors.
Stage 4S	Stage I or 2 tumor with dissemination to bone/liver/skin : - In an infant. - Good prognosis.

Risk stratification :

Low risk	High risk
• Stage I • Stage II • Stage 4S • Age of onset <18 months • Genetics : - Hyperdiploidy. - Trk A gene expression.	• Stage 3/4 • Age of onset >18 months • Genetics : - Hypodiploidy. - N-myc gene amplification. - Loss of heterozygosity (p,11q)

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Treatment :**Wilms' tumor**

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Features :usually unilateral, AKA **nephroblastoma**.

- m/c solid tumor of the kidney in childhood.
- Second m/c intra abdominal tumor in children.
- Cell of origin : **nephrogenic rest** (undifferentiated mesenchyme of the kidney).
- Age : 2 to 5 yrs.

Associations/risk factors :

1. malformations :

- **Horse shoe kidney** → 4 times risk of Wilms tumor.
- Cryptorchidism/undescended testis.
- Hypospadias.

2. Genetic syndromes :

D/t WT-1 gene (1p 13) mutation

1. **WAGR syndrome** : Wilms' tumor

Aniridia
Genitourinary anomalies
Mental Retardation

2. **Denys drash syndrome**

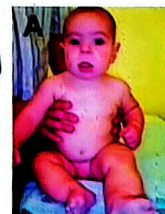
Diffuse mesangial sclerosis → Congenital nephrotic syndrome.

Association : ambiguous genitalia in males.

D/t WT-2 gene (1p 15) mutation

Beckwith-Wiedemann syndrome :

- LGA baby with hemihypertrophy
- Macroglossia.
- Omphalocele.
- Organomegaly (liver/spleen).
- Ear lobe creases.



Hemihypertrophy

Clinical features :

- Asymptomatic abdominal mass in the loin (d/t enlargement of the kidney).
- ↑ renin → Hypertension in children.
- Hematuria.

Site of metastasis : **Lung** (m/c) > lymph node > liver.

Staging:

Children oncology group (COG) staging: Post surgical staging

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Stage 1	Limited or confined to kidney, completely resectable.
Stage 2	- Extends outside the kidney margins, completely resectable. - Lymph nodes are not involved.
Stage 3	- Residual tumor after surgery confined to the abdomen. - Regional lymph node metastasis. - Tumor extension into IVC.
Stage 4	- Lymph node metastasis outside abdomen. - Hematogenous metastasis/mets to organs like lung, liver.
Stage 5	Bilateral kidney involvement.

Treatment:

